Beyond the Neighborhood: Regional Amenities in Sorting Models*

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Abstract

Equilibrium sorting models treat amenities as local; they are enjoyed equally within a geographic unit and not at all outside of it. However, many amenities, especially environmental ones, generate benefits across broad regions. Incorporating these regional amenities requires solving two problems: what is their market extent and how do benefits attenuate within it? I demonstrate that arbitrary choices defining market extent can cause order of magnitude differences in amenity valuation. To address this problem, I use a recreation demand model to construct a spatial index measuring neighborhoods' quality-adjusted exposure to a network of amenities. I demonstrate with a sorting model of Iowa and find that people have significant willingness to pay for the regional amenity of lakes. In counterfactuals, I find that intraregional heterogeneity of housing supply elasticity can cause incomplete capitalization of regional amenities, and that equal quality improvements at different sites can generate dramatically different welfare gains depending on the availability of substitutes. These findings demonstrate that accounting for the spatial extent and substitutability of public goods is essential for environmental policy.

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1 Introduction

Household location decisions have consequential impacts on adult and child welfare and later-life outcomes (Chetty et al., 2014, 2016). One factor shaping these outcomes is exposure to neighborhood amenities such as schools and air quality. Research on such amenities often treats them as club goods, with access effectively purchased through residential location. However, many goods are less geographically excludable: beaches, sports stadiums, trauma centers, or national parks can all benefit residents beyond the neighborhood they lie in. These goods are defined by what I call regionality; their value is not concentrated within a particular geography, instead it varies continuously in space. This trait has made exposure to regional goods difficult to quantify, with researchers often relying on proxies such as distance to the good, minimum thresholds to be considered as having access, or the share of nearby land occupied by the good[1]. The difficulties of regionality are particularly severe for recreational environmental goods, which often have many nearby substitutes and vary in quality. The fundamental question is one of market extent: who do we model as being excluded, and how much value do they get from access?

This paper implements a novel approach for estimating willingness to pay (WTP) for regional goods in sorting models. I use revealed-preferences data on travel decisions to estimate an amenity demand system in which households choose from a network of sites, identifying how quality and travel cost shape demand. I then summarize this system in a microfounded spatial index that measures the value of the full network at any location. When calculated for each neighborhood in a sorting model, the index measures quality-adjusted exposure to the amenities in question. Critically, this approach endogenizes market extent using revealed preferences rather than leaving it as an arbitrary decision for the researcher. It also accommodates multiple dimensions of site quality and naturally accounts for substitutability across amenities.

[1]Examples include Muehlenbachs et al. (2015); Keiser and Shapiro (2019); Greenstone and Gallagher (2008); Walsh et al. (2011); Wolf and Klaiber (2017); Gamper-Rabindran and Timmins (2013)

I apply this approach in an equilibrium sorting model of Iowa, with lakes as the amenities of interest. Lakes are a useful case study for examining regionality: they vary significantly in both quality and accessibility. Many neighborhoods have access to multiple nearby sites, while others are distant from any, making the value of access dependent on the full network of high and low-quality sites rather than distance to any given lake. Further, they are plausibly important to utility, with 65% of Iowans visiting a lake in a given year and an average of 7.8 single-day trips among those going (Wan et al., 2021). Finally, lake locations are plausibly less endogenous to contemporary residential demand than those of amenities such as hospitals or community colleges. Iowa therefore provides a useful setting to study the valuation of regional amenities while limiting the concerns of the endogenous amenity literature, (Almagro and Domínguez-Iino, 2024; Walsh, 2007).

My model is estimated on 3,500 Iowa neighborhoods across 3 year-blocks spanning 2010-2022. I use $\sim$ 650,000 observed housing sales to estimate each neighborhood's representative price, with other characteristics drawn from state and federal data. To construct the exposure index, I utilize the travel cost data from the Iowa Lakes Project, a repeated multiyear survey of Iowans' recreation habits as utilized in Ji et al. (2020, 2022). I start by demonstrating the problem of arbitrary market extent by taking a distance threshold approach: I estimate the sorting model separately for 15 different minimum distances for a neighborhood to be considered as having access to a lake. Within these specifications, the estimated willingness to pay to be within the threshold varies between \$13,000 and \$35,000. More dramatically, I show that the marginal value of any given lake, calculated as the lost surplus per household from its removal, varies sharply across these fifteen specifications. The median ratio of lowest to highest value is 12.7$\times$ with some lakes exceeding 100$\times$. This demonstrates the core difficulty regionality brings to public goods valuation: as argued by Smith (1993), the marginal value of a public good is less important than how many people we attribute that value to. Regional goods make that a difficult question to answer.

I then implement my index approach. Estimating the same model using the index in-

stead of distance thresholds, I find a WTP of $5,945 for a one standard deviation increase in lake quality-accessibility. The median lake raises per-household consumer surplus by around $5.51, with a few very high-value lakes over $100 raising the mean to $21.12. I also show that failing to account for lake accessibility in any way attenuates the coefficient on prices by 8-15%, resulting from a threat to the instrument which I describe further below. I go on to supplement the model with a supply side drawn from Baum-Snow and Han (2024), and demonstrate several implications of regionality through counterfactuals. First, I simulate the creation of Gray's Lake at its central location in Des Moines. Because of heterogeneity in supply elasticity, the demand shock to the region surrounding the lake is blunted. In particular, price increases in low-elasticity neighborhoods are attenuated by nearby high-elasticity substitutes that absorb the demand shock. This dynamic is unique to regional goods that shock multiple neighborhoods at once, as opposed to a local good change which would sharply increase demand for just one location. I further estimate two twin counterfactuals assessing the benefits of potential cleanup projects for Lakes Rathbun and Little Spirit. I model the welfare effects of equal improvements in Secchi depth, one of my dimensions of lake quality. I find that a cleanup in Rathbun, a relatively isolated lake with few nearby substitutes, delivers roughly 20 times higher welfare gains than the same improvement in Little Spirit, with many nearby substitutes. This highlights the importance of accounting for spatial substitution behavior in policy affecting regional goods.

My first contribution is to the growing spatial environmental literature, bridging insights from quantitative spatial and environmental economics (Balboni and Shapiro, 2025). The quality of regional and local environmental amenities shapes location attractiveness for household sorting decisions (Walsh et al., 2011; Wolf and Klaiber, 2017; Gamper-Rabindran and Timmins, 2013; Hamilton and Phaneuf, 2015). Separately, a longstanding quantitative spatial literature uses market access indices to summarize spatial patterns in a single term (Donaldson and Hornbeck, 2016; Ahlfeldt et al., 2015; Redding and Rossi-Hansberg, 2017). In combining these approaches, I highlight a spatial characteristic of nonexcludability in

public goods: distance limits effective access without creating a discrete boundary between access and exclusion. By estimating the spatial index on microdata rather than calibrating it from macro moments, I allow the effective market extent of a public good to emerge from the distribution of site quality, availability, and travel costs. Doing so captures rich patterns of demand substitution across sites from any given neighborhood which simple proxies cannot reflect.

My second contribution is to the literature on public goods valuation. Equilibrium sorting models have been widely used to value local public goods, including air quality (Sieg et al., 2004; Bayer et al., 2009; Tra, 2010, 2013; Hamilton and Phaneuf, 2015), open space (Walsh, 2007; Klaiber and Phaneuf, 2010), and urban heat (Klaiber et al., 2017; Borsky et al., 2024). Other studies have valued regional amenities using hedonic models Muehlenbachs et al. (2015); Keiser and Shapiro (2019); Greenstone and Gallagher (2008); Walsh et al. (2011); Wolf and Klaiber (2017); Gamper-Rabindran and Timmins (2013), of particular interest for me are Phaneuf et al. (2008); Kuwayama et al. (2022), who also use the spatial index approach to measure regional good exposure. My contribution is threefold. First, I document and address a threat regional amenities pose to the standard instrument in equilibrium sorting models, which relies on distant neighborhood characteristics to affect prices only through sorting. The issue is that housing and land-use characteristics of neighborhoods near a regional amenity can correlate with unobserved utility in more distant neighborhoods, where the amenity still provides recreational value, thus threatening the exclusion restriction. Second, I document that intraregional heterogeneity in supply elasticity causes regional goods' valuation to be undercapitalized. It is well-established that amenities are incompletely capitalized into house prices where housing supply is elastic Hilber (2017). What I find is that regional goods are subject to a second, spatial version of this problem where nearby substitute neighborhoods' elasticities also play a role, meaning hedonic estimates may be attenuated regardless of local supply elasticity. Finally, by moving from hedonic to equilibrium sorting, I soften the assumptions required to do general equilibrium analysis, particularly enabling

novel counterfactual exercises for regional goods using a supply side.

The paper continues as follows, Section 2 lays out the equilibrium sorting model, Section 3 introduces the data, Section 4 shows results and policy counterfactuals.

2 Model

The empirical strategy has two layers. The first is a residential sorting model in which agents make a discrete choice decision over different neighborhoods, each modeled as a bundle of characteristics. The second is a recreation-demand model of trips to lakes, which delivers an amenity exposure index for every location in the model, used as one of those neighborhood characteristics. I present the sorting model first, since it is the object ultimately estimated and disciplines the interpretation of everything else, followed by the housing-supply side used for general-equilibrium counterfactuals. I then show two alternative ways of measuring site exposure that do not require the travel-cost layer — indicator thresholds and continuous distance — since these are the natural benchmarks against which the index is judged in Section 4; finally I present the travel-cost model itself and show how its output becomes the index.

2.1 Sorting Model

Neighborhoods are the products in a static discrete-choice demand system over residential location. A household i chooses to buy in neighborhood j during time period t if doing so gives it the highest utility among all $L = J \times T$ available location-time-periods. Utility from choosing $l = \{j, t\}$ is

$$u_{i,l} = \alpha p_l + X_l \beta + \xi_l + \epsilon_{i,l}$$

where p_l is the prevailing home price, X_l is a vector of observed neighborhood characteristics, ξ_l is a location-time-period-specific unobservable capturing everything about j's desirability

in period t that is not in X_l, and $\epsilon_{i,l}$ is a Type-I extreme value taste shock idiosyncratic to household i. Aggregating over households with i.i.d. $\epsilon_{i,l}$ gives the standard logit market share for l,

$$s_l = \frac{\exp(\alpha p_l + X_l\beta + \xi_l)}{\sum_{c=1}^{L} \exp(\alpha p_c + X_c\beta + \xi_c)}.$$

Inversion In order to estimate the model I follow the standard two-step approach of Berry (1994) using the instrument as implemented for equilibrium sorting by Bayer et al. (2004). First, observed market shares $s_l = n_l / \sum_c n_c)$ are inverted via the contraction mapping[2]

$$\delta_l^{(k+1)} = \delta_l^{(k)} + \ln s_l - \ln \hat{s}_l\big(\delta^{(k)}\big), \qquad \hat{s}_l(\delta) = \frac{e^{\delta_l}}{\sum_c e^{\delta_c}},$$

which converges to the unique mean utility level $\delta_l = \alpha p_l + X_l\beta + \xi_l$ that makes the model exactly match observed shares in every location.

Instrument Because ξ_l is unobserved to the econometrician but is correlated with price, α cannot be recovered by ordinary least squares. Following Bayer et al. (2004), I construct an instrument that takes advantage of the sorting behavior implied by the model to satisfy relevance, using the influence of distant substitute sites for which exclusion plausibly holds. The assumption is that the availability of distant sites only affects price through sorting decisions. In particular, for each location l I construct concentric annuli[3] around the neighborhood and define Z_l as the mean of every characteristic of X_l for each neighborhood in the annuli. I further use land-use characteristics L_l which inform the difficulty of construction in other nearby neighborhoods. The above are all assumed to be correlated with ξ_l, so we

[2]With homogeneous preferences, as used throughout this paper, the standard log-inversion is equivalent to the fixed-point iteration.

[3]0–5, 5–10, 10–20, and 20–30 miles

proceed as follows: given a guess of α, α_0^*, I partial price out of the mean utility,

$$\delta_l - \alpha_0^* p_l = X_l\psi + Z_l\phi + L_l\zeta + \tilde{\xi}_l, \tag{1}$$

and estimate (ψ, ϕ, ζ) by OLS. The fitted value $X_l\hat{\psi} + Z_l\hat{\phi} + L_l\hat{\zeta}$ is an estimate of the portion of mean utility explained by l's own observed and nearby characteristics; the residual, $\tilde{\xi}_l$, is thus representative of influence originating beyond the annuli we set. Setting $\tilde{\xi}_l = 0$ and adding $\alpha_0^* p_l$ back, we have

$$\hat{\delta}_l = \alpha_0^* p_l + X_l\hat{\psi} + Z_l\hat{\phi} + L_l\hat{\zeta} \tag{2}$$

Which we substitute into the market clearing condition

$$\frac{\exp\left(\alpha_0^*\, p_l^{IV} + X_l\hat{\psi} + Z_l\hat{\phi} + L_l\hat{\zeta}\right)}{\sum_{c=1}^{L}\exp\left(\alpha_0^*\, p_c^{IV} + X_c\hat{\psi} + Z_c\hat{\phi} + L_c\hat{\zeta}\right)} = s_l \qquad \forall\, l, \tag{3}$$

And solve for the price vector p^{IV} for which predicted logit shares equal the observed shares s_l. This new price vector is then used as the instrument on price to estimate

$$\delta_l = \alpha p_l + X_l\beta + \xi_l \tag{4}$$

which recovers a new $\hat{\alpha}_1^*$. In order to mitigate the effect of the initial guess of α_0^*, the above process is repeated until convergence[4].

The instrument works because it is informed by the residual variation in $\tilde{\xi}$, which by construction can only originate from sites beyond our annuli. It is thus representative of the influence of distant site characteristics through the sorting process, distant enough that we assume no correlation between their characteristics and ξ_l. Note that this assumption is precisely what a regional amenity threatens: if a regional amenity itself is omitted or

[4]This usually happens within 5 loops

mismeasured, the resulting error is, by construction, similar for not just nearby locations but anything in the region, contaminating the instrument meant to be orthogonal to it. While including a measure of regional amenity exposure directly in X_l can keep this source of spatial correlation out of ξ_l, the cruder the exposure measure, the more of the true spatial correlation in lake access is left for ξ_l to absorb, and the more this identifying assumption is strained. I further strengthen the plausibility of this assumption in my work by including much wider annuli than is common in the literature, so as to capture only the influence on neighborhoods so far away that even regional amenities likely do not correlate.

Welfare. Given estimates $(\hat{\alpha}, \hat{\beta})$ and the recovered $\hat{\xi}_l$, the compensating variation from a change in characteristics that moves the choice set from $\{X_l\}$ to $\{X_l'\}$, holding $\hat{\xi}_l$ fixed at each location, is the standard logit consumer-surplus formula

$$\Delta CS = \frac{1}{\hat{\alpha}} \left(\ln \sum_{l=1}^{L} e^{\hat{\alpha} p_l + X_l \hat{\beta} + \hat{\xi}_l} - \ln \sum_{l=1}^{L} e^{\hat{\alpha} p_l + X_l' \hat{\beta} + \hat{\xi}_l} \right). \tag{5}$$

This is a partial equilibrium estimate measuring how much you would have to compensate the representative household to be indifferent to the new choice set. Because this welfare measure aggregates over the entire choice set, it prices in substitution: if a household's preferred location becomes less attractive relative to a close substitute as a result of a counterfactual equilibrium, ΔCS will reflect that the household can simply move.

Some counterfactuals in Section 4 instead ask what happens if households are *not* free to re-sort. Particularly, when we are interested in the partial equilibrium spatial impact of a certain amenity, we want to know how much value different neighborhoods lose from changes in their characteristics due to change in the amenity. In that case, the relevant choice set for location l collapses to a single alternative, and the compensating variation for the household(s) already living in l is simply

$$\Delta CS_l^{\text{no re-sort}} = \frac{(X_l' - X_l)\hat{\beta}}{\hat{\alpha}},$$

I use this measure whenever I want to isolate how a change in local characteristics is distributed across the spatial model without also mixing in the (separate) question of how much substitution mutes it.

2.2 Counterfactuals and the Supply Side

In order to estimate general equilibrium counterfactuals I also implement a housing supply side. This allows prices to realistically fluctuate in response to changes in amenity value and accessibility. Particularly in the case of regional amenities, having both relatively elastic and inelastic neighborhoods available causes interesting interactions in sorting because amenities are partially shared across locations.

I assume constant elasticity of supply in each Census tract, $Q_l^s = A_l \, p_l^{\eta_l}$, with tract-level elasticities η_l taken from Baum-Snow and Han (2024)[5] and the scale parameter A_l calibrated so that observed price and quantity satisfy the supply curve exactly at baseline. Given a change in characteristics, I solve for a new market-clearing price vector by iterating between (i) recomputing logit demand shares given the current price vector and (ii) adjusting each location's price in proportion to the gap between demand and supply at that price, in the spirit of the tâtonnement process in Sieg et al. (2004), until prices converge. Because demand for a regional amenity spreads across every location within reach of it rather than concentrating on a single treated neighborhood, this setup lets locations with different supply elasticities absorb different shares of the same regional demand shock — the central mechanism behind the price-capitalization results in Subsection 4.5.

2.3 Threshold and Distance Approaches

Before introducing the exposure index, I briefly introduce two measures of regional amenity exposure that have been common in the hedonic literature.

[5]Where estimates are unavailable, I simply assume perfect elasticity, since tracts without estimates are typically low-population and remote. They generally do not affect counterfactuals significantly

Indicator thresholds. A simple approach lets a location "have" the amenity if the nearest site falls within a given distance d of it. Defining dist_l as the minimum distance to a site, the threshold enters the sorting model as $\mathbb{1}[\text{dist}_l \leq d]$. In order to investigate the stability of estimates to d, I estimate the full sorting model separately for fifteen values of d from 1 to 15 kilometers. This measure's biggest weakness is its arbitrary definition of market extent, allowing for no one outside the radius exposure and everyone inside equal exposure when realistically the function is continuous. Ideally, these issues would cancel out in such a way that the larger the radius, the lower the average value from exposure per-resident but the more residents, such that the marginal valuation of any given site is similar across specifications, thus making the arbitrary threshold innocuous.

Continuous distance. This approach is most commonly used for unique amenities such as distance to a city's central business district. Still, it can be adapted to a context with multiple sites as distance to the nearest site, along with the square to allow for nonlinearity. These two measures simply enter as

$$X_l \ni \left(\text{dist}_l,\ \text{dist}_l^2\right).$$

This avoids the arbitrary cutoff but introduces the concern that there is no limit to market extent: everyone gets some exposure to some amenity. Critically, both the threshold and distance measures are purely geometric: they encode where a site is but nothing about it. A site's estimated value under either measure is therefore driven entirely by a single common valuation among all sites, and how many households happen to live near it, with no way to discount low-quality sites from high-quality ones. This is a particularly costly omission in a setting with real variance in quality, since it means removing a low-quality site is treated identically to removing a high-quality one at the same location.

2.4 Travel Cost Model and the Amenity Exposure Index

In my attempt to address the above concerns over market extent, value attenuation, and differences in quality, I attempt to measure amenity exposure more directly. I accomplish this through a quality-accessibility index built from a recreation demand model of trips to the amenity in question, following the logic of Phaneuf et al. (2008) and Kuwayama et al. (2022). On any of a fixed number of choice occasions in a year, a household living in location l visits one of J sites. The (non-random) value of visiting site j from l in year y is

$$a_{j,l,y} = \kappa_{j,y} + \tau_y \, tc_{l,j,y},$$

where $\kappa_{j,y}$ is an alternative-specific constant (ASC) capturing site j's average desirability in year y, later to be decomposed into site characteristics, and $tc_{l,j,y}$ is the round-trip travel cost between l and j,

$$tc_{l,j,y} = 2\left(c_y \cdot d_{l,j} + \tfrac{1}{3}\,\text{time}_{l,j} \cdot \text{wage}_{l,y}\right),$$

where c_y[6] is per-mile travel costs, $d_{l,j}$, the driving distance between l and j, and the opportunity cost of travel time is valued at one-third of the local wage multiplied by $\text{time}_{l,j}$, the driving time, following the standard travel-cost convention (Ji et al., 2020, 2022; Zhang et al., 2024; Cheng and Wan, 2026). τ_y is estimated, and reported throughout, as a raw coefficient rather than a magnitude to be subtracted; it is negative (Table 5), so higher travel cost lowers utility as expected. Adding a Type-I extreme value taste shock gives visitation probabilities

$$P(\text{visit } j \mid l) = \frac{e^{a_{j,l,y}}}{\sum_{k=1}^{J} e^{a_{k,l,y}}}$$

[6] from AAA's published year-specific driving-cost estimates

Writing $n_{i,j,y}$ for the number of trips household i took to site j in year y, out of the fixed number of choice occasions, the log-likelihood is

$$\ell\ell(\{\kappa_{j,y}\}, \tau_y) = \sum_i \sum_{j=1}^{J} n_{i,j,y} \ln P(\text{visit } j \mid l_i),$$

I maximize this separately for each survey year, recovering one ASC per site-year along with an annual travel-cost parameter τ_y. The ASCs are themselves the object of interest for recovering preferences over site characteristics: in a second stage, I regress $\kappa_{j,y}$ on observed site characteristics X_j,

$$\kappa_{j,y} = \gamma X_j + \varepsilon_{j,y},$$

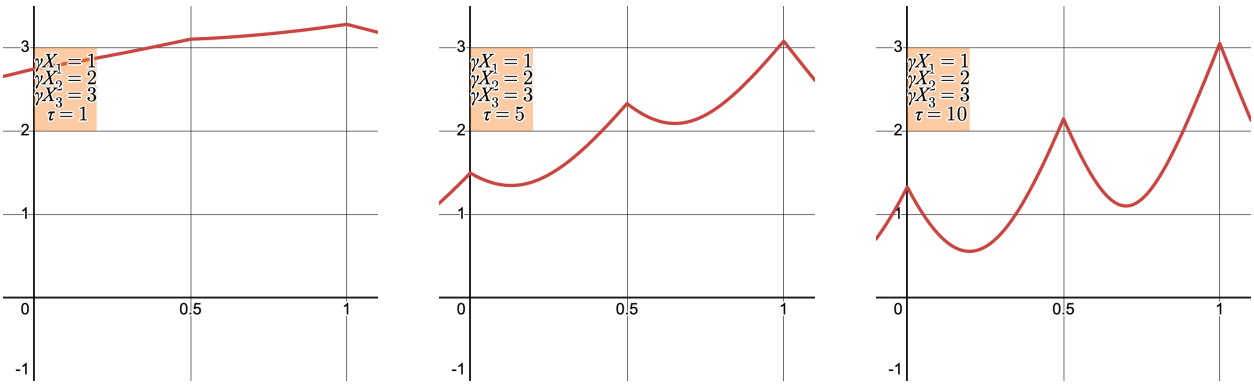
by OLS. While this could also be done separately by year, I find no evidence that preferences for these site characteristics change over time, so to simplify I use a pooled result[7]. The fitted $\hat{\gamma}$ (reported in Table 6) is what lets the index be recomputed for a site whose characteristics change, or for a hypothetical site that does not exist in the estimation sample, by predicting $\hat{\kappa}_j = \hat{\gamma} X_j$ rather than requiring κ_j to have been separately estimated for it. For the same reason, I use a single $\hat{\tau}$, averaged across all seven survey years, rather than the year-specific $\hat{\tau}_y$: the index needs one consistent travel-cost sensitivity applicable to every grid cell and year-block in the sorting model, not seven different ones tied to the recreation survey's own calendar.

The quality-accessibility index for location l is the expected value of the best available recreation opportunity from l,

$$\hat{A}_l = E\big[\max(u_{i,1,l}, \ldots, u_{i,J,l})\big] = \ln\left(\sum_{j=1}^{J} e^{\hat{\kappa}_j + \hat{\tau}\, tc_{l,j}}\right),$$

[7] Ji et al. (2020) finds that preferences do shift for certain characteristics over time. Since the eventual welfare object here is net present value of all future amenity flows, it doesn't make sense to use different preference parameters each year

the standard logit inclusive value, evaluated at the pooled $(\hat{\tau}, \hat{\gamma})$ rather than at any single survey year's estimates. $\hat{A}_l$ rises in the quality of nearby sites and falls in the travel cost of reaching them. Removing or changing X_j for one site changes $a_{j,l}$ for that site at every location simultaneously, in proportion to each location's travel cost to it. Thus, a counterfactual index recomputes $\hat{A}_l$ for every neighborhood in the model, and the size of $\hat{\tau}$ determines how far away that change will be meaningfully felt. Figure 1 shows how the index varies for the same set of amenities given higher or lower τ, it is the key parameter dictating spatial attenuation of amenity exposure.



(a) $\tau = 1$ (b) $\tau = 5$ (c) $\tau = 10$

Figure 1: The value of A_l across space when $\gamma X_k = k$ and amenities are located at 0, .5, and 1. Note that the index captures both the value of being near any amenity and the value of being closer to higher-quality amenities

3 Context and Data

I implement the model using Iowa's lakes as the regional amenity of interest. Lakes are a natural setting for this approach because they fail every simplifying assumption that makes localized-amenity sorting models tractable: their recreation benefits are not confined to a single neighborhood, they vary along several dimensions of quality, many locations can reach more than one of them, and access is highly uneven across the state.

3.1 Recreation Demand Data

The travel-cost model requires origin-destination trip data together with lake characteristics; the construction of travel cost itself, and its estimation, are described in Subsection 2.4. Trips come from the Iowa Lakes Project, a repeated survey of Iowa households' recreation behavior conducted by Iowa State University in 2002–2005, 2009, 2014, and 2019. Respondents report the number of single-day trips taken to each of Iowa's significant lakes, pictured in Figure 2, over the preceding season; following Ji et al. (2020) I treat each household as facing 52 choice occasions per year. Driving distance and time are measured between the household's (or, when calculating the index at the grid-cell level, the grid cell's) centroid and each lake; the per-mile driving cost is taken from AAA's *Your Driving Costs* series, and the opportunity cost of time from the household's self-reported income (or, for the grid cell, its median income), imputed to an hourly wage. Lake characteristics are drawn from the Iowa Department of Natural Resources and the Iowa State Limnology laboratory. Of particular interest is Secchi depth, a measure of visual lake clarity that estimates how far into the water one can still see. Various impurities such as phytoplankton from algal blooms or Phosphorus from sediment runoff cause lower Secchi depth and give water an unclean look. Summary statistics for lake characteristics are in Table 1.

Statistic	N	Mean	SD	Min	Max
Log area	892	4.795	1.687	2.303	9.852
Secchi depth (m)	892	1.175	0.941	0.087	8.100
Boat ramp (0/1)	892	0.857	0.348	0	1
State park (0/1)	892	0.380	0.486	0	1

Table 1: Lake characteristics, pooled across the 7 travel-cost survey years (892 site-years, 123–139 distinct lakes per year)

3.2 The Gridded EIF

A core concern with choosing how to define neighborhoods in the sorting model is spatial granularity. My variable of interest, lake accessibility, should not meaningfully vary within

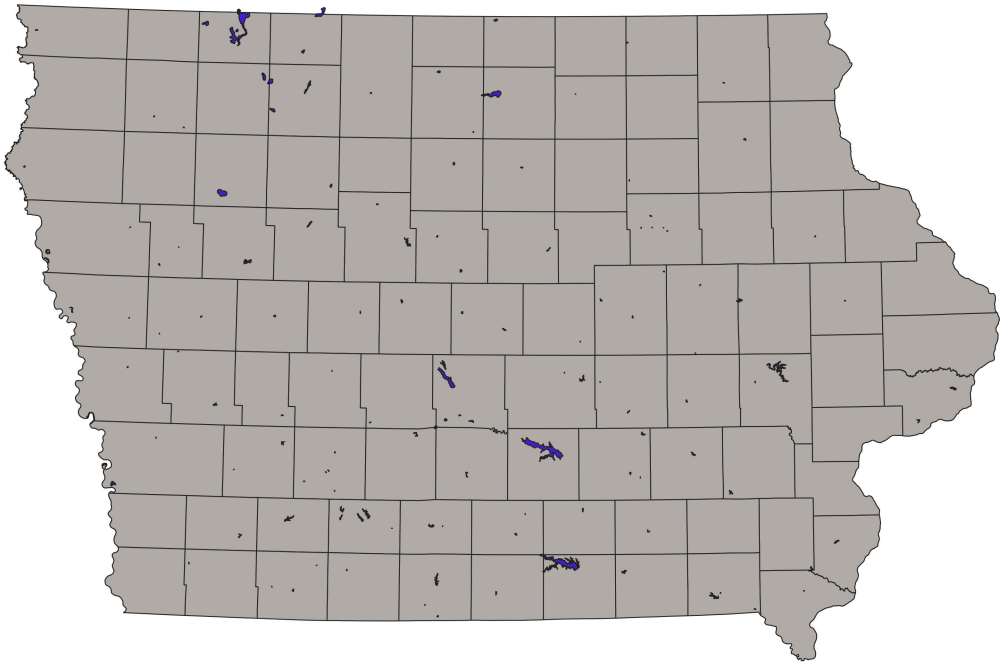


Figure 2: The Lakes Considered in the Recreation Demand Study

the chosen geography. Census tracts and even block groups are a poor fit for this requirement: their boundaries are drawn so as to keep population per unit roughly similar, so their physical size varies enormously. A rural Iowa block group can span tens of square miles, while a Des Moines block group can be a few blocks. A single lake index value assigned to an entire rural block group could easily average over travel costs that differ by an hour's drive from one edge of the unit to the other, which is precisely the within-unit heterogeneity the index is meant to resolve.

I instead use the Census Bureau's Gridded Environmental Impacts Frame (Gridded EIF), a privacy-protected data product that reports population counts by race, income, age, and sex on a uniform $0.01^\circ \times 0.01^\circ$ latitude-longitude grid (roughly 0.7 by 0.9 kilometers in Iowa) covering the entire United States back to 1999. Quantitative spatial economics has increasingly turned to fine, population-independent grids rather than administrative units precisely because market-access and amenity-exposure measures are sensitive to the within-

unit heterogeneity that tracts and block groups paper over (Donaldson and Hornbeck, 2016; Redding and Rossi-Hansberg, 2017), but that literature has generally built its own custom grids from scratch, raster population counts, or nighttime lights, each with its own coverage and privacy tradeoffs. The Gridded EIF offers a nationally standardized, privacy-protected alternative with demographic detail at the same resolution, assembled directly from Census and IRS administrative records rather than imputed from satellite imagery. To my knowledge, this paper is the first to use it as the geography for a sorting model. Because grid cells are a fixed-size geography defined independently of population density, political boundaries, or the location of the amenity being studied, a cell's travel cost to any given lake is far more homogeneous within the cell than within a tract or block group. Iowa is covered by over 100,000 such cells, but we will restrict to the 3,557 with adequate housing sales data (Subsection 3.3). These are disproportionately the state's more populated cells, containing roughly 75% of Iowa's population and 85% of its observed housing sales. In addition to providing geometry, the gridded EIF provides my measure of per-cell % white, median income, and PM2.5 exposure, summarized in Table 2.

Statistic	N	Mean	SD	Min	Max
% White	10,671	87.19	12.72	5.71	100.00
Median income ($10,000s)	10,671	8.07	5.96	−1.28	59.33
PM 2.5 ($\mu g/m^3$)	10,671	7.87	1.15	5.66	11.57

Table 2: Gridded-EIF-derived characteristics across the 10,671 modeled grid-cell-by-year-block locations. A handful of very low-income-decile, low-population cells produce the negative minimum for median income; this reflects the underlying synthetic income imputation in sparsely-populated cells rather than a data error.

3.3 Housing Sales and Prices

Home prices are built from CoreLogic's deed-transfer records, which cover Iowa property sales as far back as the early 20th century. I keep arm's-length transactions with a recorded price of at least $5,000, dropping split-parcel sales and same-day duplicate records at the same parcel, and restrict to the three year-blocks used in estimation (2010–2013, 2014–2017,

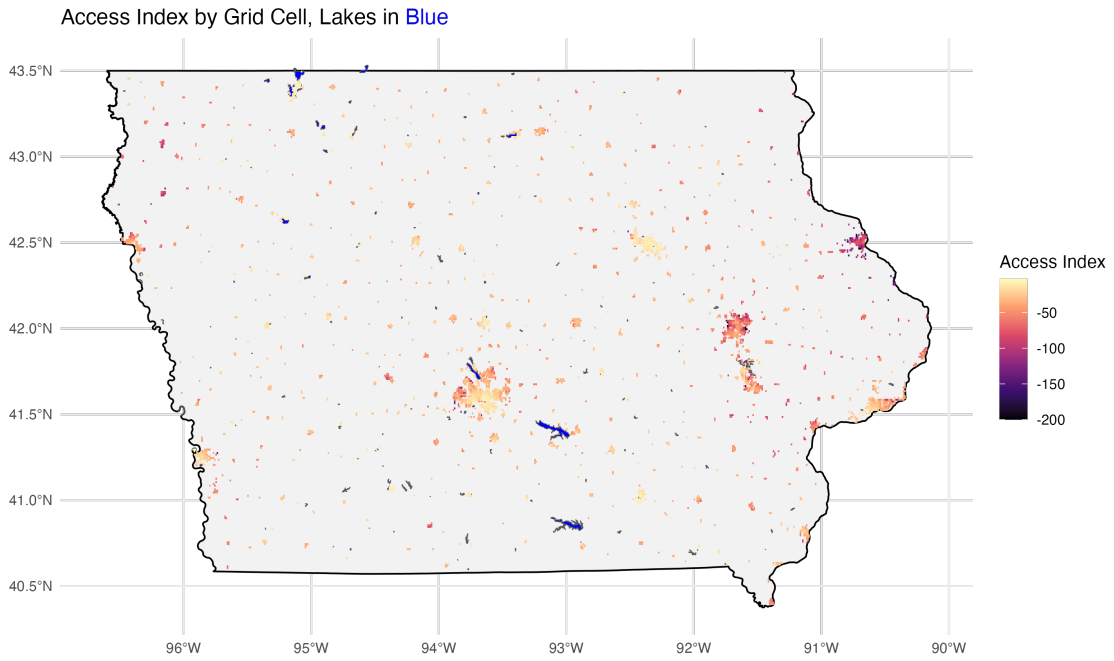


Figure 3: Map of all used grid cells, colored by index value

2018–2022), following Klaiber and Phaneuf (2010) and Bayer et al. (2009) in splitting the sample this way to balance transaction density against within-block price appreciation. This leaves 643,011 individual sales matched to a parcel- or block-level coordinate.

A grid cell's representative price for a year-block should not simply be the average sale price observed there for all the years in the year-block, since sale prices drift within a year-block due to appreciation over the years a block spans. I first deflate every sale to 2010 dollars using BLS CPI-U annual averages, then, for each cell-by-year-block product h, recover a representative log-price π_h from

$$\ln P_h^i = \pi_h + y_t^i \theta_t + \varepsilon_h^i,$$

where P_h^i is the (CPI-deflated) price of the ith sale assigned to product h and y_t^i are calendar-year indicators. This is estimated as a single two-way fixed-effects regression across all products and all sales at once, with π_h recovered as the product fixed effect and θ_t a common

year effect shared across every cell, following the price-adjustment approach of Bayer et al. (2009). $\rho_h = \exp(\widehat{\pi_h})$ is the representing price entering the sorting model.

A cell-by-year-block product is only as reliable as the number of sales backing it. Rather than impute a price for thinly-transacted cells (these cells often have almost no homes in them to start with), I require at least 5 sales in a given year-block for a grid cell to be assigned its own product in that block. Because the model is already static, I do not want certain locations to only be available at certain year-blocks, so any grid cell that cannot clear this bar in all three year-blocks is dropped from the choice set entirely. This leaves 3,557 grid cells, or 10,671 total grid-cell-by-year-block locations, with the sales-per-product counts summarized in Table 3.

Statistic	N	Mean	SD	Min	Max
Price (ρ_h, 2010\$)	10,671	130,065	78,950	11,471	853,604
Sales per product	10,671	49.95	49.37	5	380

Table 3: Recovered grid-cell-by-year-block prices and the number of sales backing each. The minimum of 5 is the binding cutoff described above, not an approximation of it.

3.4 Other Neighborhood Characteristics

As discussed in Subsection 2.1, we require data on land use across the state of Iowa. Typically, studies such as Klaiber and Phaneuf (2010) and Bayer et al. (2004) draw on local data about land use that has detailed description on whether land is used for particular densities of residential, intensities of commercial use, and more. With my area of study being the full state of Iowa, no such locally-assembled land use data exist, and combining multiple local sources may leave me with inconsistent definitions of land use. Instead, I draw on the National Land Cover Database (NLCD), which classifies every 30-meter pixel in the country into one of several land cover categories, including four separate intensities of developed land (open, low, medium, and high development) alongside open water, forest, shrub/scrub, grassland, pasture/hay, cultivated crops, and wetlands. This is less-detailed than is ideal, but given over 80% of Iowa's land is used for agriculture, it is likely sufficient.

I also compute median structural housing characteristics — square footage, bedrooms, bathrooms, and lot size — per grid-cell-by-year-block product from the same CoreLogic transaction records used for price, taking the within-product median for all sales. Finally, I construct an indicator for whether a grid cell's centroid lies within 10 miles of Iowa's state border, motivated by the concern that the index, built only from Iowa's own lakes, cannot see any recreational lakes just across the border which households near the border plausibly have access to.

Statistic	N	Mean	SD	Min	Max
Median sqft (100s)	10,671	17.08	5.89	4.60	56.23
Median bedrooms	10,671	3.06	0.57	1.00	20.00
Median bathrooms	10,671	2.19	0.87	1.00	8.00
Median lot size (acres)	10,671	0.33	0.51	0.00	18.62
On border (0/1)	10,671	0.227	0.419	0	1
% open water (0–1mi)	10,671	3.20	7.86	0.00	71.65
% developed, open (0–1mi)	10,671	11.16	6.61	0.00	38.98
% developed, low (0–1mi)	10,671	21.97	11.56	0.00	68.12
% developed, medium (0–1mi)	10,671	13.42	11.68	0.00	66.74
% developed, high (0–1mi)	10,671	3.20	4.71	0.00	49.42
% forest (0–1mi)	10,671	6.49	9.59	0.00	69.39
% shrub/scrub (0–1mi)	10,671	0.00	0.03	0.00	1.20
% grassland (0–1mi)	10,671	0.45	2.43	0.00	51.81
% pasture/hay (0–1mi)	10,671	8.01	9.37	0.00	64.96
% cultivated crops (0–1mi)	10,671	29.47	26.01	0.00	95.35
% wetlands (0–1mi)	10,671	2.41	4.60	0.00	51.49

Table 4: Structural housing characteristics, border proximity, and nearest-ring (0–1 mile) land use shares across the 10,671 modeled locations. Land use percentages are shares of land area, 0–100.

4 Results

4.1 The Trouble with Thresholds

I first estimate the sorting model with lake access measured as an indicator for being within d kilometers of the nearest lake, separately for every integer d from 1 to 15. Each threshold requires a full re-estimation of the model because these are fifteen different definitions of

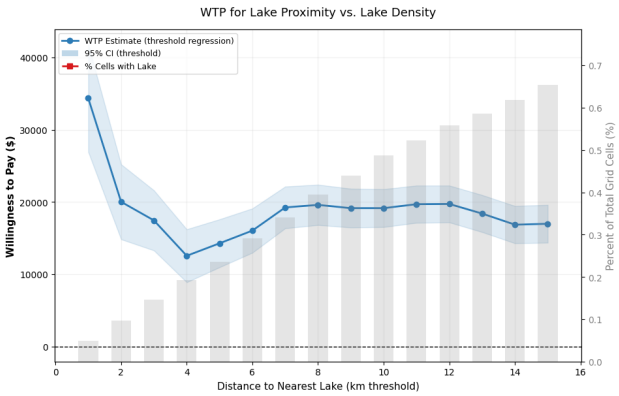


Figure 4: Willingness to pay for lake access, defined with 15 different distance cutoffs

what counts as "having a lake nearby." The implied WTP for lake access is estimated as $\beta_{\mathbb{1}(d_{l,lake}<d)}/\alpha$ for each of the models. An ideal measure of lake access would be invariant to this choice in one specific sense: widening the threshold waters down the definition of "accessible," so per-person WTP should fall smoothly as more, more-distant households are included, with the two effects roughly offsetting so that the *total* value attributed to lakes is stable. Figure 4 shows this benchmark fails. Per-person WTP does fall as expected out to 4km, but then it rises again and stabilizes after 7km, even as the treated population continues to grow from 20% to 65%. Total value derived from lakes is therefore not stable across specifications — it increases purely because of how the researcher chose to define accessibility, which is the opposite of what the benchmark requires. This is concerning, but to see the problem more clearly we must consider the exercise of valuing individual lakes.

For a given threshold d and a given lake, I compute the consumer surplus lost from removing that lake. This is accomplished by removing the lake, re-estimating whether each neighborhood falls within the threshold under the resulting counterfactual distances, and using Equation 5 to price the change for a representative agent. If the threshold measure is accurately capturing the value of lake accessibility, this quantity would ideally be roughly *constant* across the fifteen specifications: the watered down definition of access roughly cancels out with the additional number of people who have access such that a researcher's arbitrary choice of cutoff does not change how valuable a given lake is. Unfortunately, this is

not the case. At many low thresholds, lakes are trivially worthless simply because they are not within the distance cutoff of any of my grid cells, this is an uninteresting artifact of the model, so Figure 5 therefore plots the minimum non-zero value against the maximum across the fifteen specifications for each lake. Even restricting to non-zero comparisons, the ratio between the largest and smallest implied value has a median of 12.7× and can exceed 100×. In other words, arbitrary decisions made by the researcher can generate orders of magnitude different estimations of value. Figure 6 illustrates the same point not just using minimum and maximum, tracing out the full fifteen-threshold profile for six representative lakes spanning the value distribution: none of the six profiles is flat, most aren't even monotonic.

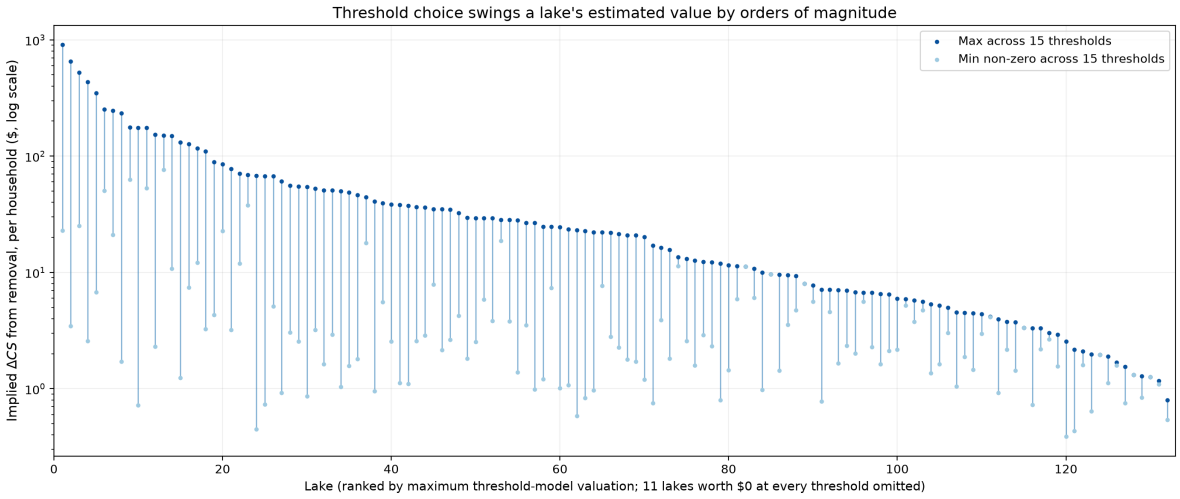


Figure 5: Minimum non-zero and maximum lost consumer surplus from removing each lake, across fifteen separately-estimated distance-threshold specifications (1–15km), ranked by each lake's maximum implied value. 11 lakes worth $0 at every threshold are omitted.

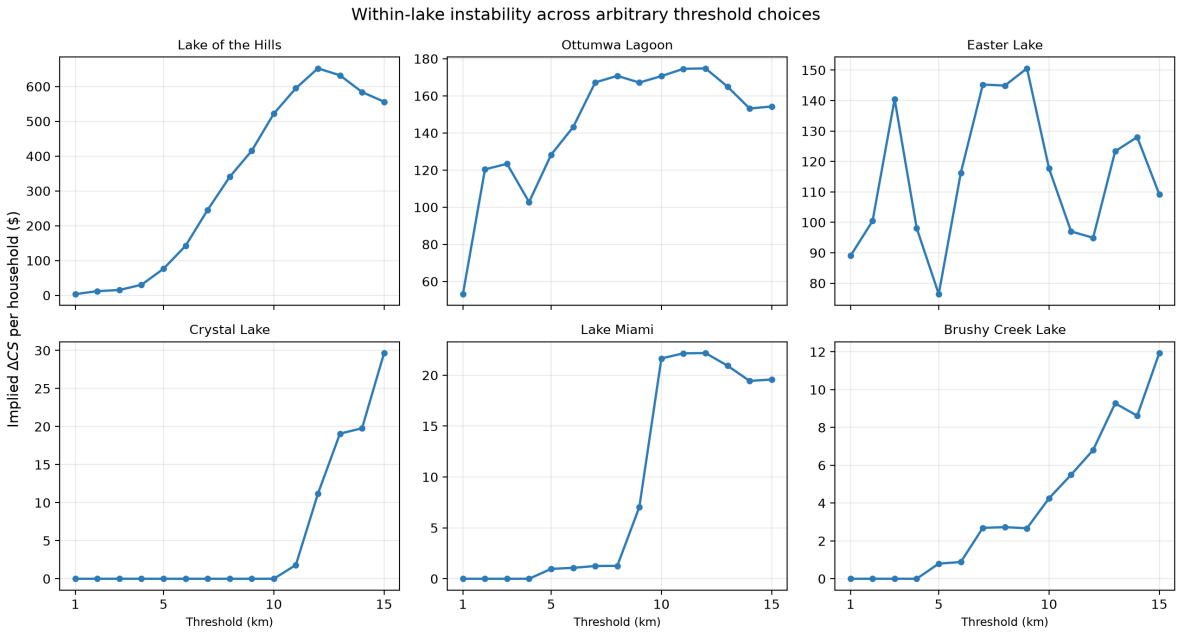


Figure 6: Threshold-by-threshold implied value for six representative lakes spanning the value distribution. Most profiles are neither flat nor smoothly monotonic, and no two share the same shape.

4.2 The Index Approach

I begin by maximizing the site-choice log-likelihood of Subsection 2.4 separately for each of the seven survey years, recovering one travel-cost parameter τ_y and one ASC $\kappa_{j,y}$ per site. Table 5 reports τ_y together with its asymptotic standard error. τ drifts substantially across years; since the welfare object estimated here is meant to be the net-present value of all future amenity flows, rather than a snapshot tied to any one survey wave, I use the average of τ_y across all seven years (as described in Subsection 2.4) for the index calculation in every year-block. Appendix A shows this pooling choice is not consequential for the headline WTP estimate: re-building the index and re-estimating the full model under each individual year's τ_y instead of the pooled mean moves WTP by at most a few percent.

Year	τ_y (SE)	Mean κ_j	SD κ_j	Sites
2002	−1.780 (0.010)	−3.772	1.124	123
2003	−1.559 (0.008)	−3.937	1.139	123
2004	−1.449 (0.008)	−3.964	1.065	123
2005	−1.697 (0.009)	−3.762	1.125	123
2009	−1.869 (0.009)	−3.776	1.034	123
2014	−2.771 (0.015)	−3.760	1.054	138
2019	−2.841 (0.021)	−3.977	1.244	139

Table 5: First-stage travel-cost model results by survey year. τ_y and its standard error come from the inverse Hessian of the maximum-likelihood site-choice model at convergence; mean and SD of κ_j summarize the recovered site ASCs for that year.

The $\kappa_{j,y}$ recovered above are not used directly; the index needs to be recomputable for a site whose characteristics change or that does not exist in the estimation sample at all, which requires decomposing $\kappa_{j,y}$ into the underlying site characteristics via the second-stage regression of Subsection 2.4, $\kappa_{j,y} = \gamma X_j + \varepsilon_{j,y}$. I pool all 892 site-year observations into a single regression rather than fit γ separately by year, on the same premise as earlier that while preferences may shift over time, we would prefer to model them as constant. Table 6 tests directly whether $\hat{\gamma}$ meaningfully shifts over time by adding a full set of year fixed effects and finds that regression coefficients are essentially unchanged. I move forward using the pooled model without fixed effects as $\hat{\gamma}$.

	(1)	(2)
Secchi depth	0.225 (0.021)	0.227 (0.021)
Log area	0.436 (0.017)	0.436 (0.016)
State park	0.238 (0.058)	0.238 (0.058)
Boat ramp	0.289 (0.088)	0.287 (0.087)
Year FE	No	Yes
R^2	0.574	0.581
N	892	892

Table 6: Second-stage decomposition of $\kappa_{j,y}$ into site characteristics, pooled across all 892 site-year observations, with and without a full set of year fixed effects. Robust standard errors in parentheses. The year fixed effects are jointly significant ($F = 2.53$, $p = 0.020$) but leave every characteristic coefficient essentially unchanged, which is what licenses pooling them into a single $\hat{\gamma}$.

With these simplifications we ensure that any change in $\hat{A}_l$ across year-blocks is not due

to varying preferences over time. I do allow variance within cell across year-blocks due to travel cost itself: the per-mile driving cost c_y[8] and the local wage used to value travel time[9], both of which enter $\hat{A}_l$ directly. This is done with the idea that there will be consistent changes in travel cost over time due to rising wages and changing transportation prices that are worth modeling. With this, $\hat{A}_l$ is computed as in Subsection 2.4, using $\hat{\tau}$, the fitted $\hat{\kappa}_j = \hat{\gamma} X_j$ for every one of the state's 143 lakes, and each cell's own block-specific travel cost to each of them. Their distribution is pictured in Figure 7, and the map of values by grid cell is pictured in Figure 3. The resulting index is standardized to mean zero and unit standard deviation[10] before entering the sorting model as a single characteristic in X_l. Table 7 reports the full set of estimated coefficients.

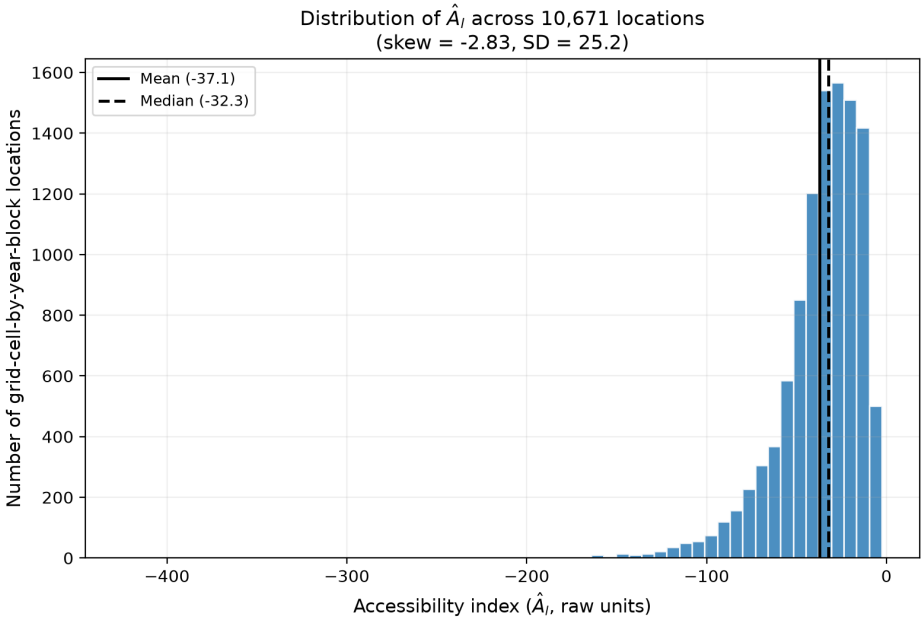


Figure 7: Distribution of the accessibility index $\hat{A}_l$ across all 10,671 modeled grid-cell-by-year-block locations.

[8]The median of AAA's published rate over the years making up that block

[9]Also the median for each cell among the years in the block

[10]These are affine transformations of a term measured in utility and thus do not affect the economic interpretation

Variable	Coefficient	SE
Price ($10,000s)	−0.1487***	(0.0049)
Access index (SD)	0.0884***	(0.0152)
Median income ($10,000s)	0.1050***	(0.0056)
PM 2.5	−0.1852***	(0.0086)
% White	0.0004	(0.0009)
Sqft (100s)	−0.0045	(0.0038)
Bathrooms	0.5501***	(0.0276)
Bedrooms	0.0757***	(0.0286)
On border	0.0494*	(0.0275)

Table 7: Second-stage sorting model results, access index specification. *, **, *** denote significance at the 90%, 95%, and 99% level. Land use coefficients are suppressed.

Dividing the index coefficient by the price coefficient gives a willingness to pay of $5,945 for a one standard deviation increase in the index. Using this coefficient, I can price the lost consumer surplus to the representative household under the counterfactual removal of each individual lake. As in Subsection 4.1, these are partial equilibrium estimates: the loss in surplus to a representative household allowed to re-sort across the full choice set in response to the lake's removal, with prices held fixed[11]. This gives a right-skewed distribution of lake values across the state's 143 lakes, with a median of $5.51 and a mean of $21.12 per household, the gap driven by a small number of high-value sites against a long tail of lakes valued near zero.

A $5,945 willingness to pay for a one-standard-deviation move might seem hard to square with a $5.51 median lake value, but the two are consistent once two things are taken into account. First is that due to the skew of $\hat{A}_l$, a one standard deviation increase can move a neighborhood from the 44th to the 95th percentile of exposure, so most of the distribution sits well within a single standard deviation of the mean and only a small number of locations are actually being moved by that much. Further, we must recall that these are marginal valuations, a single lake being lost may not move the total index value much, if we were to remove several in the same region the results would be more dramatic. The magnitude is

[11]In full equilibrium, prices also adjust, and — as Subsection 4.5 and Subsection 2.2 show — can absorb enough of the shock that the resulting change in consumer surplus is substantially smaller.

also not dissimilar from the results Kuwayama et al. (2022) found when including recreation access in their hedonic regression in the context of Tampa Bay.

We can now benchmark the index approach against the threshold and distance approaches. Figure 8 compares the index specification to the continuous-distance specification, with each lake's per-household ΔCS ranked by its value under the index model. Figure 9 repeats this comparison against the threshold models, plotting each of the fifteen threshold specifications' per-household values. Overall, the distance model tends to value sites more highly than the index approach does, though not uniformly so. The threshold models show a similar tendency (setting aside the many lakes they mechanically set to zero value), especially for the lower-value lakes.

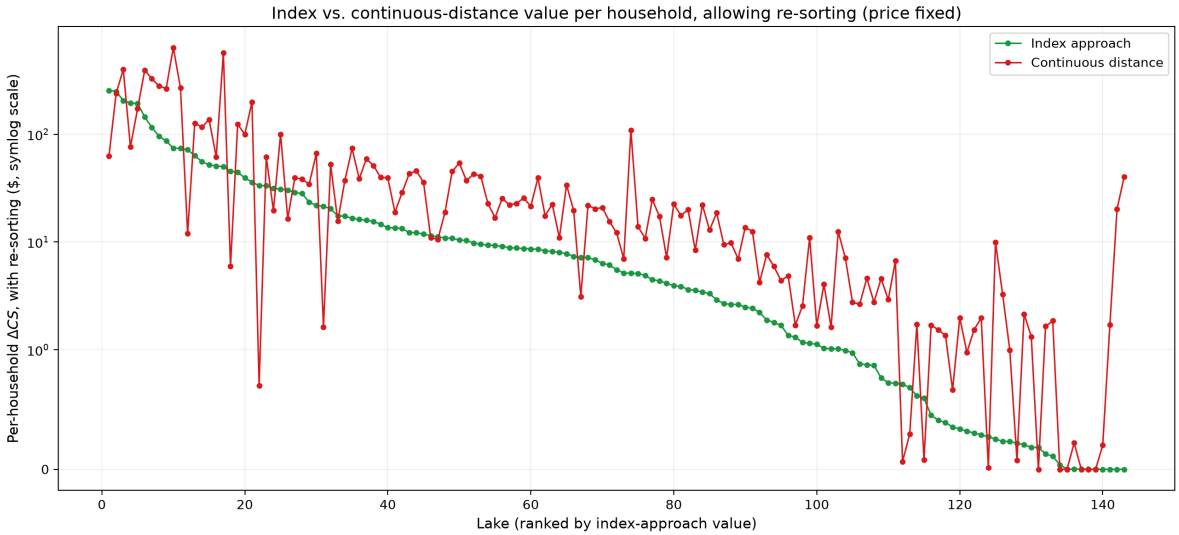


Figure 8: Per-household ΔCS for each lake under the index and continuous-distance approaches, allowing re-sorting at fixed prices, ranked by the index approach's valuation. The index values decline smoothly because of ranking.

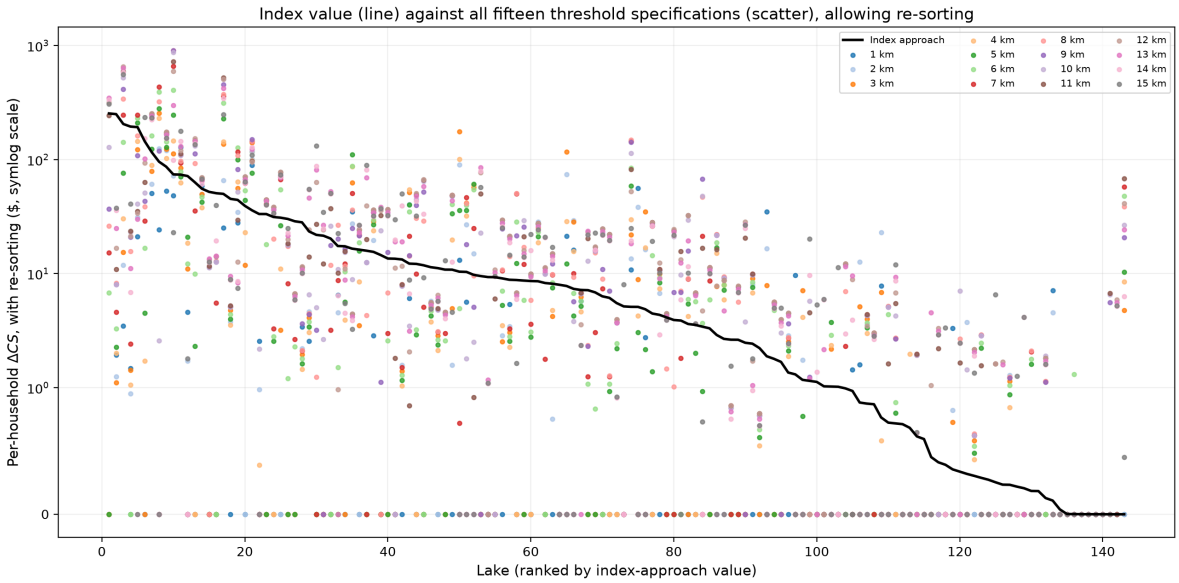


Figure 9: Per-household ΔCS for each lake under the index and 15 different threshold models, allowing re-sorting at fixed prices, ranked by the index approach's valuation. The index values decline smoothly because of ranking.

What is driving these disparities is the subject of the next section: the approaches that only take distance into account ignore the other things that drive a lake's value to a given location.

4.3 How Site Value Varies with Distance

The clearest illustration of why modeling exposure using simple heuristics is so difficult is to look directly at how a lake's value at a given location changes with distance. For each lake, and for every grid cell in the state, I compute that cell's change in partial equilibrium value upon lake removal. Figure 10 plots this per-cell value against distance from the lake for four representative lakes, together with the binned mean trend. None of the four patterns is well approximated by a simple decay function, nor by a binary cutoff for inside and outside a threshold. In fact, value is not even monotonic as a function of distance. This non-monotonicity is not noise, it is the direct result of substitution through the underlying index. Two cells at equal distances but opposite directions from the same lake can have different

valuations of it, because one may have immediate access to a second lake nearby while the other has only the first. Non-monotonicity arises because a cell can be further from a lake but still value it more due to isolation. Fitting any single cutoff or continuous, monotonic function of distance to this pattern — across 143 lakes whose value-by-distance profiles take none of the same shape — cannot recover it, no matter how flexibly the function is specified, because the object being approximated is simply not a function of distance alone.

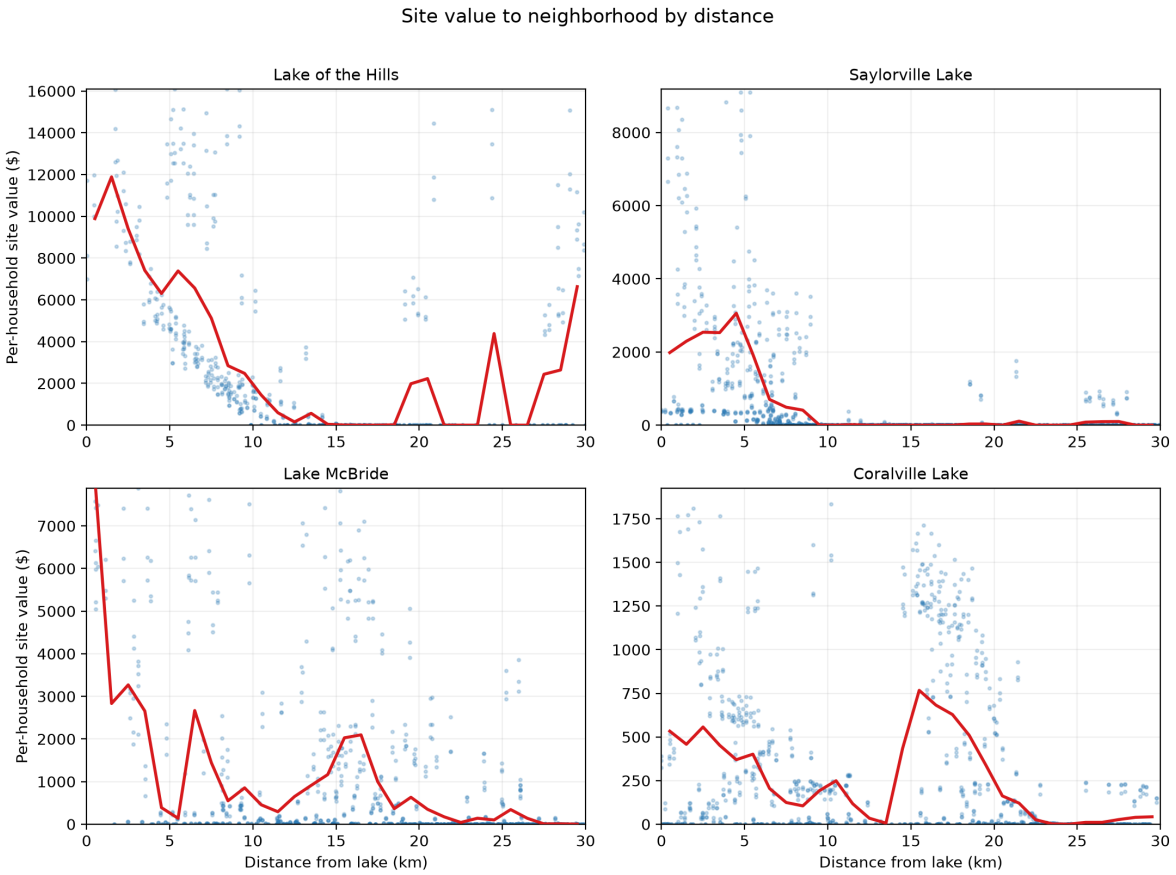


Figure 10: Per-household site value (no re-sorting, not weighted by population) against distance from the lake, for four lakes with continuous grid-cell coverage out to 30km. Blue points are individual grid cells; the red line is a binned mean.

4.4 Omitted Regional Amenities and Instrument Validity

The ring-based instrument of Subsection 2.1, following Bayer et al. (2004), relies on an exclusion restriction: the characteristics of distant locations used to construct p_{iv} must be

uncorrelated with a given location's own unobserved utility ξ_l. This restriction is harder to defend in the presence of a regional amenity. If lake access is not measured directly, it goes into ξ_l, but the lakes themselves are plausibly located outside the 5 mile rings standard in the literature. This means if any neighborhood characteristics correlate with being near lakes, including the controls for land use themselves, the exclusion restriction is violated. The problem here is precisely that the good is regional: it is far enough away to not be controlled for as contributing to local utility, but close enough that it still affects utility.

I test this directly by re-estimating the full instrument-construction and 2SLS procedure of Subsection 2.1 without any measure of lake-access in the own-cell or ring-averaged characteristics used to build p_{iv}. Table 8 reports the resulting price coefficient $\hat{\alpha}$. Dropping the access index from its own specification attenuates $\hat{\alpha}$ by 8.4 percent[12]; dropping distance or the threshold indicator attenuates it by even more. Attenuation is towards zero, the expected direction for a valuable amenity being loaded into price. As a comparison, removing median number of bedrooms — a characteristic with its own significant, well-measured effect on price, but which does not plausibly inform utility for distant locations — from the index specification moves $\hat{\alpha}$ by only 0.35 percent. The instability is therefore not a generic consequence of dropping any one regressor; ordinary omitted variables are well-addressed by the instrument, but regional amenities are not. I stress that this occurs even under my rings which control for land use characteristics up to 30 miles out, well beyond what is common in the literature.

Specification	$\hat{\alpha}$	SE	% shift if dropped
No lake-access measure	−0.1361	(0.0047)	—
Access index	−0.1487	(0.0049)	−8.4%
Distance to nearest lake	−0.1586	(0.0053)	−14.2%
Threshold (≤8km)	−0.1552	(0.0052)	−12.3%

Table 8: Price coefficient $\hat{\alpha}$ under different lake-access specifications, each re-estimated with the full instrument-construction procedure. % shift if dropped is the change in $\hat{\alpha}$ from that row's specification to the no-lake-access baseline.

[12]I confirm via a 500 draw paired bootstrap at the grid cell level that $\hat{\alpha}_{\text{with}} - \hat{\alpha}_{\text{without}} = 0.0126$ is nonzero, the bootstrap SE is .0037 with a 95% confidence interval of $[-0.0214, -0.0061]$

This sensitivity is also not the result of instrument weakness leading to large swings. Table 9 reports the first-stage F-statistic and partial R^2 of p_{iv} with and without the access index; both specifications easily clear conventional weak-instrument thresholds. Together these results show that regional amenities are a real threat to the exclusion restriction for the ring instrument in sorting models.

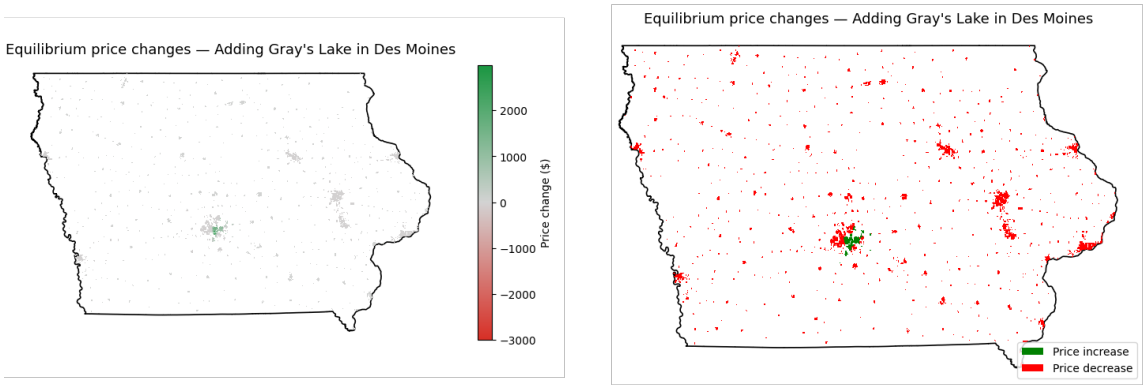
Specification	First-stage F (robust)	Partial R^2
With access index	1,985.6	0.2247
Without access index	1,781.2	0.2249

Table 9: First-stage strength of p_{iv}, with and without the access index. F is the robust (HC1) Wald statistic on p_{iv} in the regression of price on p_{iv} and the remaining exogenous controls; partial R^2 is net of those same controls.

4.5 Counterfactuals: Creating Gray's Lake

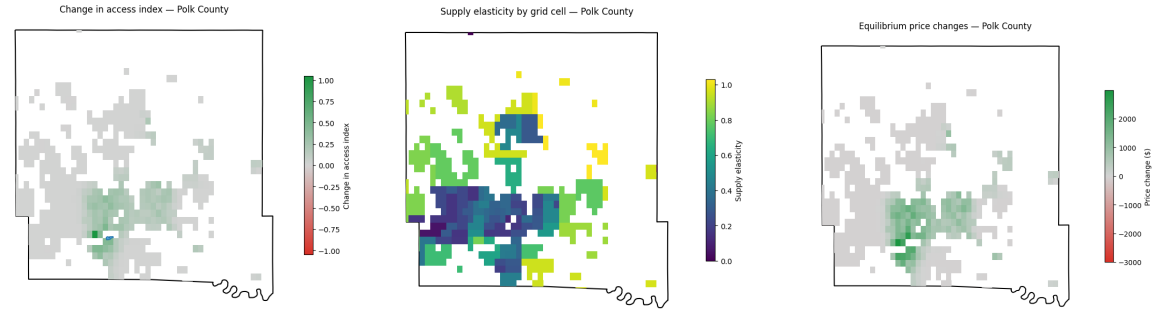
I estimate a general equilibrium response to the creation of a major amenity, illustrated with Gray's Lake, an urban lake in downtown Des Moines and one of the highest-value lakes in the model. Creating it in partial equilibrium raises per-household surplus by \$95.66 (aggregate market welfare ≈\$51.0 million). The shock to demand is concentrated around Des Moines and its immediate suburbs. Closing the model with the tract-level supply elasticities as per Subsection 2.2 shows how this demand shift translates into prices; as is visible in Figure 11, prices rise sharply in Des Moines and its immediate suburbs, while falling modestly everywhere else.

More interesting is the role of supply elasticity in shaping these price changes. Because there is meaningful heterogeneity in supply elasticity within Des Moines, high-elasticity neighborhoods respond to the demand shock partially through increases in the quantity of housing supplied rather than through price alone. Because these neighborhoods are themselves substitutes for the less-elastic neighborhoods, they effectively dampen the price increase there by absorbing some of the excess demand that would otherwise fall on the more inelastic neighborhoods. Figure 12 zooms in on Polk County, containing Des Moines, and



(a) Change in GE Prices (b) Binary change in GE Prices

Figure 11: Effect on Prices of Creating Gray's Lake Statewide



(a) Change in Index Value (b) Supply Elasticity (c) Change in Price

Figure 12: Effect of Gained Lake Exposure on Prices, Moderated by Elasticity

visualizes how the shock to the index does not translate directly into a change in price, with certain neighborhoods seeing far less of an increase than others.

While these results are simply summary statistics of the mechanical output of the estimated model, I demonstrate this relationship via simple regression, limited to cells in Polk County:

$$\Delta\text{Price}_l = \beta_0 + \beta_1 \Delta\text{Access}_l + \beta_2 \text{Elasticity}_l + \beta_3(\Delta\text{Access}_l \times \text{Elasticity}_l) + \beta_4 \overline{\text{Elasticity}}_{-l} + \beta_5(\Delta\text{Access}_l \times \overline{\text{Elasticity}}_{-l}) + \epsilon_l$$

where $\overline{\text{Elasticity}}_{-l}$ is the mean elasticity of all other cells within 5 miles of l. The results are in Table 10 and show that β_5 is negative: the more elastic a cell's neighbors, even conditional on its own elasticity, the less it is affected by the demand shock.

Variable	Coef.	SE
Constant	−73.48	9.64***
ΔAccess	5,966.08	173.33***
Elasticity	−9.21	6.28
ΔAccess × Elasticity	−444.06	205.39**
$\overline{\text{Elasticity}}_{-l}$	73.90	20.24***
$\Delta\text{Access}_l \times \overline{\text{Elasticity}}_{-l}$	−1,906.59	448.36***
R^2	0.991	
N	1,377	

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 10: Equilibrium price response to Gray's Lake's creation, Polk County cells only. Robust (HC1) standard errors.

This finding is more than the standard concern that amenities are incompletely capitalized into price where supply is elastic (Hilber and Mayer, 2009). That is present here and evidenced by the significant negative interaction on β_3. The novel issue here is that the elasticity in *other* locations further prevents complete capitalization even in highly inelastic cells because the amenity shock is common across them. This is a further dampening effect unique to regional amenities, such that hedonic models may struggle to accurately value regional amenities even when accurately measured and in a low supply-elasticity context.

4.6 Counterfactuals: Lake Cleanups and Nearby Substitutes

Because water clarity, which I measure via Secchi depth, is one of the characteristics valued in the travel-cost model, the index approach can also price a quality improvement to an existing lake, not merely its removal. I use this to evaluate a federally-funded project to reduce Phosphorus loading in Lake Rathbun in southern Iowa, budgeted at $12 million, and to demonstrate the importance of taking the full spatial distribution of amenities into account. Phosphorus reduction is not itself a characteristic in the travel-cost model, but

it is one of the primary determinants of Secchi depth, so I simulate a range of potential counterfactual Secchi depths the cleanup may result in, recomputing the quality-accessibility index for each, and compute the compensating variation for each, essentially the total value of the cleanup.

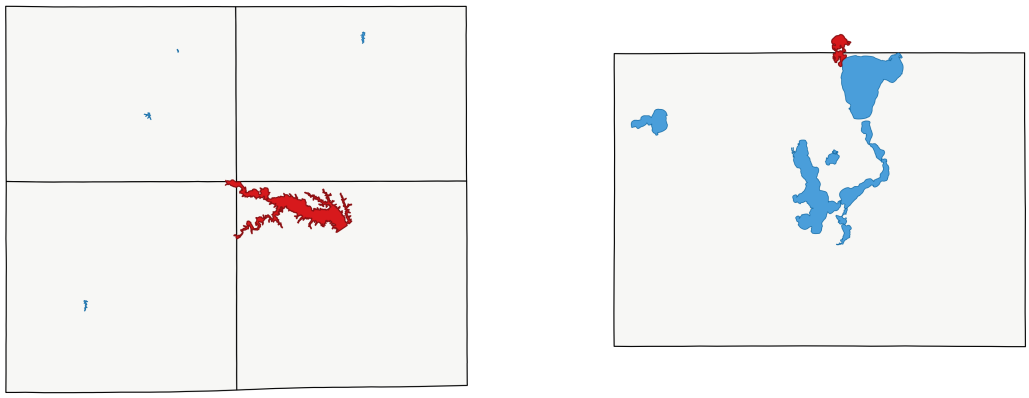
I compare Rathbun against Little Spirit Lake in northern Iowa, chosen to hold water quality similar while varying substitute availability and size sharply. Table 11 summarizes the two lakes: both are similarly turbid at baseline, but they differ starkly in isolation. Rathbun's nearest lake is 29.4km away, and no lake lies within 15km of it at all. Little Spirit sits adjacent to Big Spirit Lake and has another four lakes within 15km, including one with a Secchi depth of 6.26 meters, one of the clearest in the state. Figure 13 maps both lakes and their county neighbors.

Statistic	Rathbun Lake	Little Spirit Lake
Secchi depth, mean (m)	0.78	0.82
Secchi depth, range (m)	0.48–1.23	0.29–1.25
Lake size (ln acres)	9.31	6.43
State park	Yes	No
Boat ramp	Yes	Yes

Table 11: Lake characteristics for the two cleanup counterfactuals. Secchi depth statistics are computed across the seven travel-cost survey years.

I simulate improvements over baseline up to 8 meters, which would make both lakes cleaner than any in the state. Figure 14 displays the results in terms of total social surplus[13]. Roughly doubling Rathbun's Secchi depth to 1.5 meters improves social welfare by $30,000, while doing the same for Little Spirit only improves welfare by $1,300. The larger the simulated improvement, the greater the level gap, at 3 meters Rathbun improves surplus by $95,000, Little Spirit only $5,000.

[13]This multiplies the representative household's compensating valuation by the number of households in the model.



(a) Rathbun Lake (Appanoose, Lucas, Monroe, and Wayne Counties)

(b) Little Spirit Lake (Dickinson County)

Figure 13: The two cleanup lakes (red) and all other lakes in the depicted regions (blue).

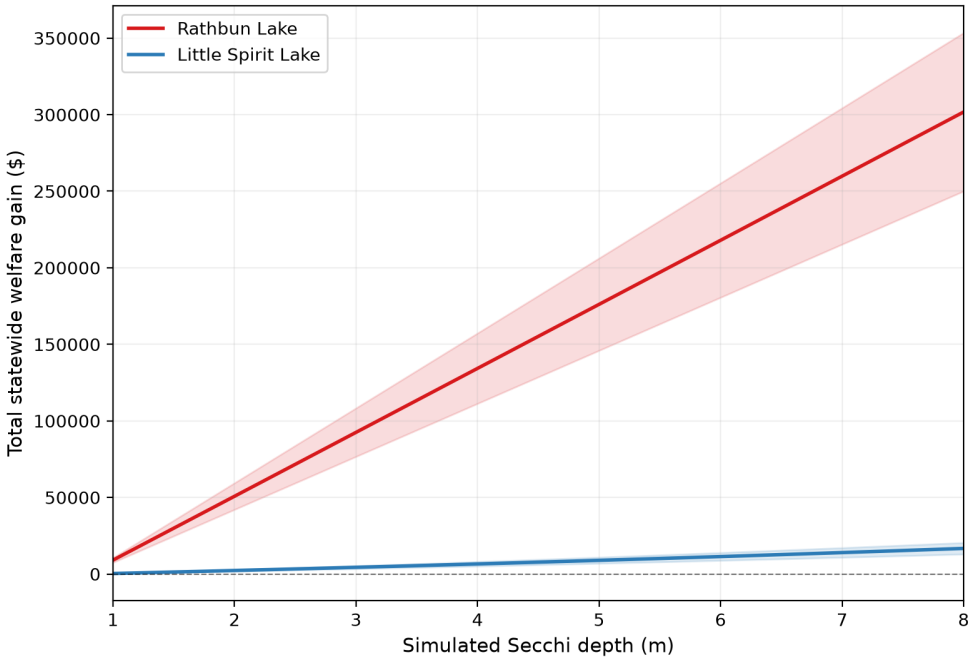


Figure 14: Simulated total statewide welfare gain from improving Secchi depth at Rathbun Lake and Little Spirit Lake, relative to each lake's estimated baseline. Shaded bands are ±1 SE.

This gap in cleanup value is partially the result of substitute availability and partially due to quality. Little Spirit lake having many high-value substitutes nearby means improve-

ments in lake quality mostly affect welfare through second order decisions: some people now substitute to this lake instead of its larger neighbors. But for Rathbun, improvements are first-order: everyone taking lake trips in the region is already visiting this one lake, so their experiences are all improved, this is in addition to some second-order effects from inducing new people to make lake trips. For a similar reason, quality matters. Rathbun is both larger and in a state park, this means it has a higher baseline visitation for first-order improvements to improve upon. The net result is that the exact same cleanup causes orders of magnitude differences in welfare improvement. This exercise would not be feasible under the heuristic approaches as they do not account for quality, but even if they were, they could not capture the importance of this substitution behavior.

To briefly address the policy itself, against a $12 million project cost, even Rathbun's most generous welfare estimate falls well short of justifying the expenditure using this valuation alone. However, it is worth pointing out that the index will not capture all potential benefits of such an improvement. It may partially capitalize things that are correlated with lake exposure, such as sediments running off into local drinking water, but would miss the value of ecosystem services generated by a cleaner lake, local and nonlocal agricultural spillovers, or effects on downstream water quality.

5 Conclusion

Many of the amenities researchers care about — lakes, parks, hospitals, transit networks, broadband — do not respect that boundary, and I show that bending the standard model to fit them anyway, by picking a threshold or a distance-decay function and hoping it is not doing too much work, is not a safe simplification. The threshold and continuous-distance approaches I benchmark against generate value estimates that differ by orders of magnitude depending on researcher-chosen specification details that have no principled answer, and the discrepancies are large precisely for the counterfactuals that matter most for policy.

The index I propose resolves this by endogenizing value attenuation directly: accessibility is not assumed to follow any particular functional form, it falls out of an actual recreation-demand model estimated on real trip data, and quality and distance enter the same travel-cost framework on equal footing rather than being bolted together after the fact. This buys more than a better point estimate. It buys counterfactuals the heuristic approaches cannot support at all: pricing a hypothetical quality improvement at every location in the state simultaneously, as in the Rathbun and Little Spirit comparison, or closing the model with a supply side to show how regional demand shocks are absorbed unevenly across space, as in the Gray's Lake exercise. Both results depend on substitution patterns that a threshold or a monotonic distance function cannot represent by construction, and both turned out to matter quantitatively: whether a lake has nearby substitutes changed the value of an identical water-quality improvement by roughly an order of magnitude, and heterogeneous supply elasticity meant that regional amenities can be incompletely capitalized into price even in markets that look supply-constrained on their own.

The paper also makes an econometric point that is easy to miss if the index is treated only as a measurement convenience: omitting a regional amenity does not just bias its own coefficient, it threatens the exclusion restriction that the sorting literature's standard instrument relies on, because a regional amenity's spatial footprint is exactly the kind of neighbor-to-neighbor correlation that instrument is supposed to be free of. I show this bias is real, not merely a theoretical concern, and that it attenuates estimated price sensitivity specifically — not a generic consequence of omitting any one characteristic.

This analysis has real limits worth stating plainly. The welfare estimates throughout are lower bounds: the index prices local, access-correlated use value well, but it has nothing to say about downstream water quality, non-residential or agricultural spillovers, or the non-use value a cleaner lake might hold for people who never visit it. Preferences are also assumed homogeneous throughout, and amenity quality is treated as exogenous to where people choose to live.

Both of these last two assumptions are natural next steps. Allowing heterogeneous preferences — for instance, letting the marginal utility of lake access vary by income, household composition, or other observables, rather than imposing a single β for the whole state — would let the model speak to distributional questions this version cannot: who actually captures the welfare gains from a cleanup, and whether the households most affected by a regional amenity are the ones currently best served by it. Treating amenity quality as endogenous is the more structurally ambitious extension: water quality responds to land use and population pressure in the same region whose sorting behavior the model is trying to explain, so a cleaner treatment would close the loop between where households sort and how that sorting itself shapes the amenities they are sorting around.

A Robustness to the Pooled Travel-Cost Parameter

Subsection 4.2 pools the travel-cost parameter τ_y across all seven survey years into a single $\hat{\tau}$ used to build the accessibility index in every year-block, rather than letting it vary by year, on the grounds that the welfare object of interest is a net-present-value flow rather than a snapshot tied to one survey wave. Because τ_y itself drifts substantially across years (Table 5), this pooling is a modeling choice rather than a data-driven necessity, so I check directly whether it matters: I rebuild the accessibility index separately under each individual year's $\hat{\tau}_y$ in place of the pooled mean, holding every other step of the index construction fixed, and re-estimate the full instrument-construction and 2SLS procedure of Subsection 2.1 from scratch under each resulting index. Table 12 reports the implied willingness to pay for a one-standard-deviation increase in the index under each of the eight specifications.

Specification	$\hat{\tau}$	WTP	SE	t
Main model (pooled)	−1.995	\$5,945	\$1,003	5.93
Year 2002	−1.780	\$6,035	\$1,002	6.03
Year 2003	−1.559	\$6,151	\$1,000	6.15
Year 2004	−1.449	\$6,223	\$998	6.23
Year 2005	−1.697	\$6,075	\$1,001	6.07
Year 2009	−1.869	\$5,997	\$1,002	5.98
Year 2014	−2.771	\$5,754	\$1,007	5.72
Year 2019	−2.841	\$5,742	\$1,007	5.70

Table 12: Willingness to pay for a one-standard-deviation increase in the access index, re-estimated under each year's own travel-cost parameter $\hat{\tau}_y$ in place of the pooled mean used in the main specification. SE and t-statistics come from the delta method applied to the full instrument-construction and 2SLS procedure under each specification.

WTP ranges from \$5,742 to \$6,223 across all eight specifications, a band of about ±4 percent around the pooled headline estimate of \$5,945, and every specification remains significant at $t \approx 5.7$–6.2. There is a mild, monotonic pattern — the early years with weaker decay in travel cost imply slightly higher WTP, while the steeper $\hat{\tau}$ estimated in 2014 and 2019 implies slightly lower WTP — but nothing close to overturning the result. The pooling choice in Subsection 4.2 is not doing meaningful work on its own.

B Robustness of Individual Lake Values to the Pooled Travel-Cost Parameter

Appendix A shows the pooled-τ choice barely moves the single headline WTP figure. That check only looks at one summary number, so I repeat it for every one of the 141 lakes with a valid fitted ASC: I rebuild the index under each of the eight τ specifications and recompute the full lake-removal welfare exercise of Subsection 4.2 from scratch under each one, giving every lake eight separately-estimated values rather than one. Table 13 summarizes the resulting distribution of lake values under each specification.

Specification	Median	Mean	CV	Corr. w/ pooled	Rank corr.
Main model (pooled)	\$6.12	\$21.42	2.092	1.00000	1.00000
Year 2002	\$6.22	\$21.75	2.091	0.99997	0.99982
Year 2003	\$6.35	\$22.18	2.089	0.99985	0.99913
Year 2004	\$6.42	\$22.46	2.087	0.99973	0.99878
Year 2005	\$6.26	\$21.90	2.090	0.99994	0.99960
Year 2009	\$6.17	\$21.60	2.091	0.99999	0.99991
Year 2014	\$5.90	\$20.70	2.095	0.99986	0.99929
Year 2019	\$5.89	\$20.66	2.095	0.99984	0.99925

Table 13: Distribution of per-household lake-removal values across all 141 lakes, re-estimated from scratch under each of the eight τ specifications. "Corr. w/ pooled" and "Rank corr." are the Pearson and rank correlations, across lakes, between each specification's values and the pooled-τ main specification's.

Not only is the distribution's shape essentially unchanged across specifications — median, mean, and coefficient of variation all move by a percent or two — the rank correlation with the pooled specification exceeds 0.999 in every year. The τ-pooling choice is not just failing to move the average lake's value, it is failing to reshuffle which lakes are high- or low-value at all.

Figure 15 makes the same comparison Figure 5 makes for the threshold specifications, plotting each lake's minimum and maximum value across the eight τ specifications, ranked by maximum value. Unlike the threshold case, no lake is ever exactly zero under the index, so nothing needs to be dropped to put this on a log scale.

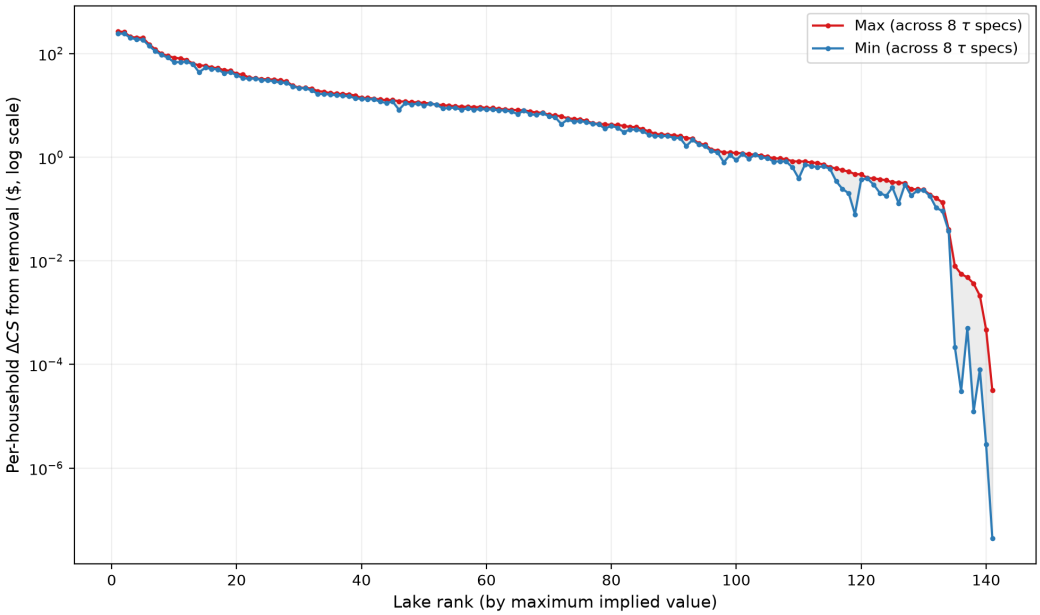


Figure 15: Minimum and maximum per-household lake-removal value across the eight τ specifications, for each of 141 lakes, ranked by maximum value. Log scale.

The median ratio between a lake's maximum and minimum value across specifications is 1.09×, against 12.7× for the threshold approach's within-lake disagreement. The raw maximum ratio, 725.6×, looks alarming until its source is identified: it belongs to lakes worth a few thousandths of a cent per household to begin with, where any ratio of two near-zero numbers is noise rather than signal, the same failure mode that motivates restricting Figure 5 to non-zero threshold values in the first place. Restricting to lakes worth at least $1 per household — 103 of the 141, and the only ones a policymaker would plausibly act on — the worst-case ratio across all eight specifications falls to 1.46×. The index's values are not just stable on average; they are stable lake by lake, for every lake large enough to matter.

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